LABORATORY WORK 6

JOIN

Database Management Systems

Task 1 — Airline flights

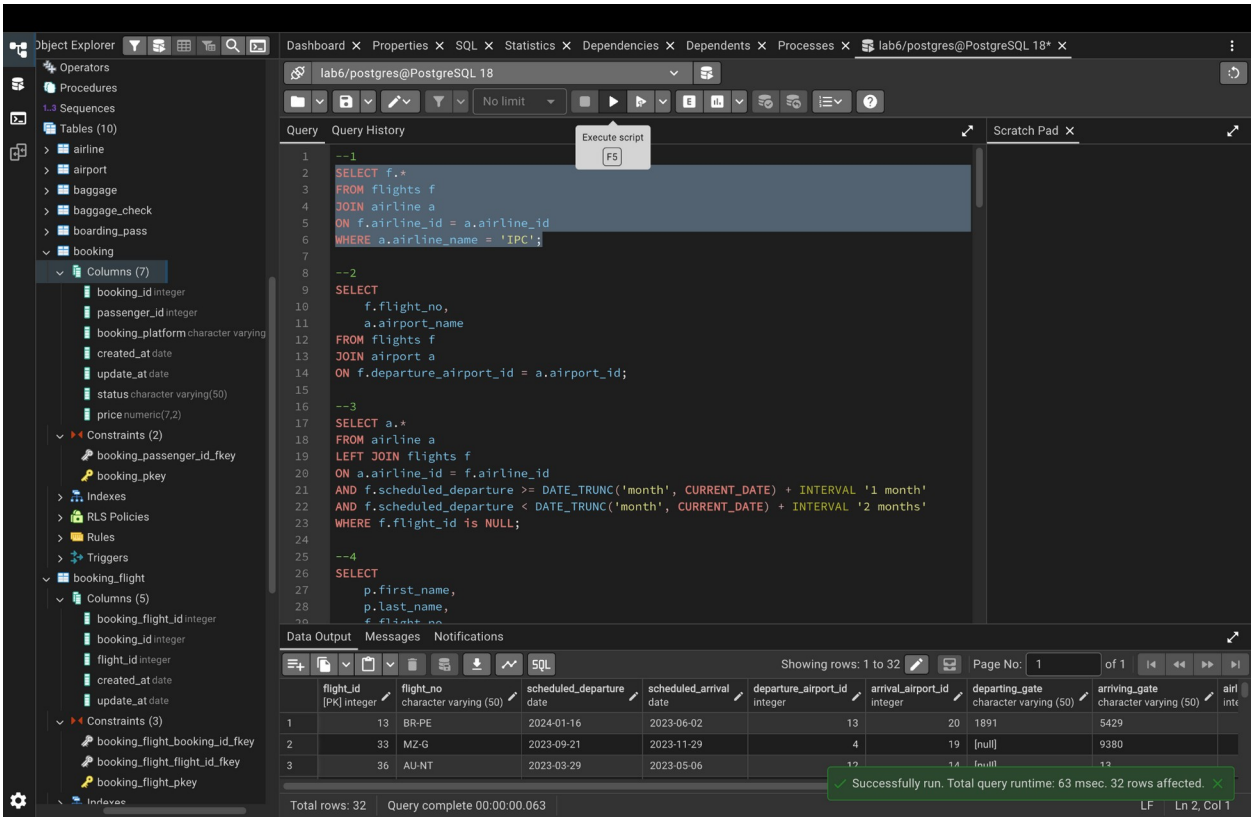
Explanation: Flights of a specific airline (IPC) are selected by joining the flights and airline tables on airline_id.

SQL Query

```
SELECT f.*
FROM flights f
JOIN airline a
ON f.airline_id = a.airline_id
WHERE a.airline_name = 'IPC';
```

Result: The query was executed successfully. 32 rows were returned.

Screenshot — Task 1



Task 2 — Flights with departure airports

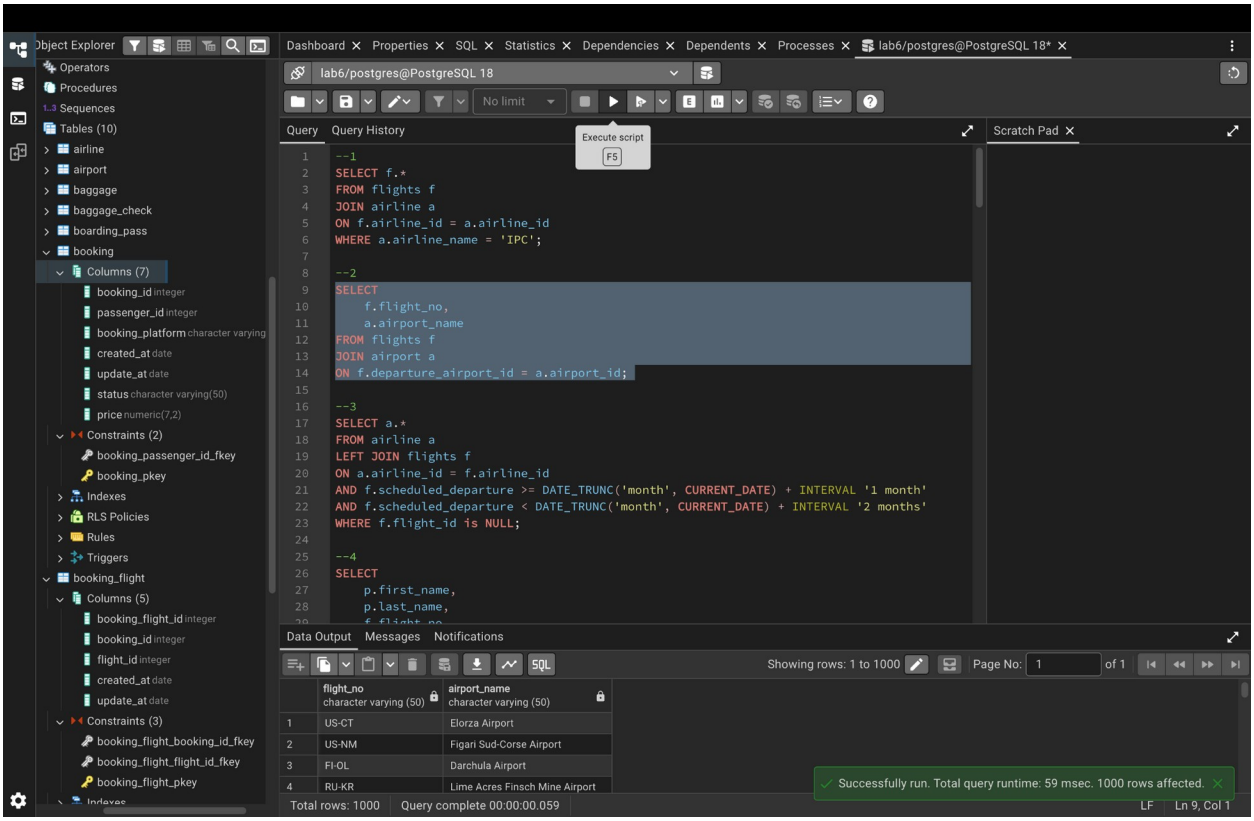
Explanation: Each flight is shown together with the name of its departure airport. The flights table is joined with airport on departure_airport_id.

SQL Query

```
SELECT
    f.flight_no,
    a.airport_name
FROM flights f
JOIN airport a
ON f.departure_airport_id = a.airport_id;
```

Result: The query was executed successfully. 1000 rows were returned.

Screenshot — Task 2



Task 3 — Airlines without flights next month

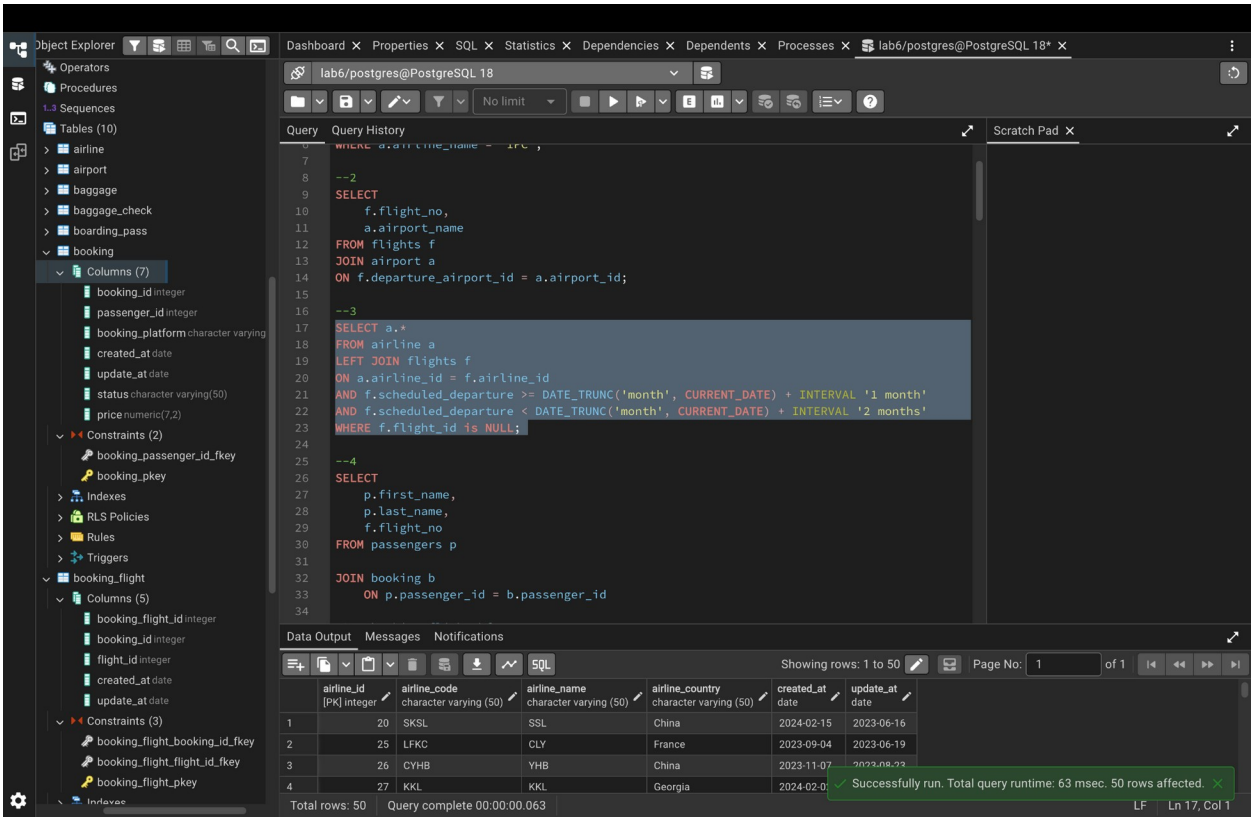
Explanation: A LEFT JOIN is used to match airlines with flights scheduled for the next month. Airlines with no matching flight (f.flight_id IS NULL) have no flights next month.

SQL Query

```
SELECT a.*
FROM airline a
LEFT JOIN flights f
ON a.airline_id = f.airline_id
AND f.scheduled_departure >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
AND f.scheduled_departure < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
WHERE f.flight_id is NULL;
```

Result: The query was executed successfully. 50 rows were returned.

Screenshot — Task 3



Task 4 — Passengers on a specific flight

Explanation: Passengers, bookings, booking_flight and flights are joined to list the passengers of flight US-CT.

SQL Query

```
SELECT
    p.first_name,
    p.last_name,
    f.flight_no
FROM passengers p

JOIN booking b
    ON p.passenger_id = b.passenger_id

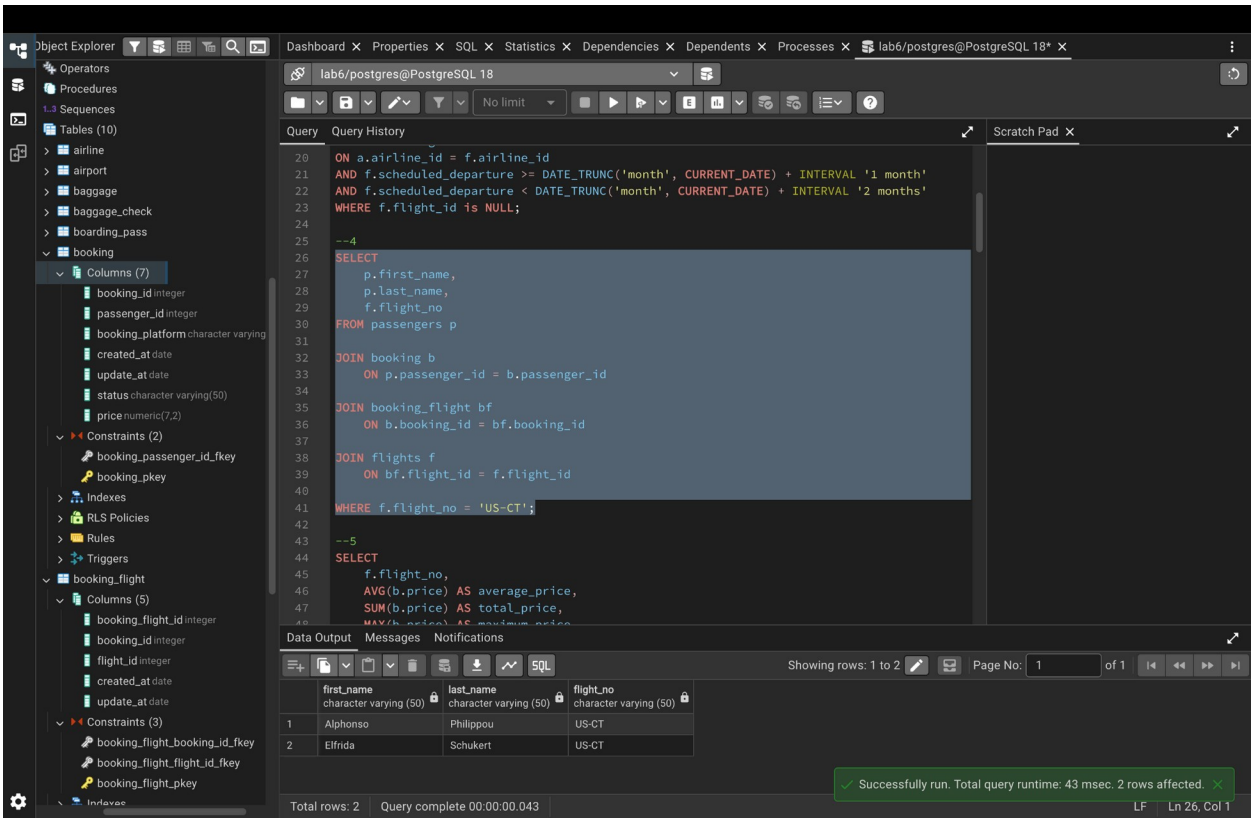
JOIN booking_flight bf
    ON b.booking_id = bf.booking_id

JOIN flights f
    ON bf.flight_id = f.flight_id

WHERE f.flight_no = 'US-CT';
```

Result: The query was executed successfully. 2 rows were returned.

Screenshot — Task 4



Task 5 — Ticket price statistics per flight

Explanation: For each flight the average, total, maximum and minimum ticket price are calculated with AVG, SUM, MAX and MIN and grouped by flight number.

SQL Query

```
SELECT
    f.flight_no,
    AVG(b.price) AS average_price,
    SUM(b.price) AS total_price,
    MAX(b.price) AS maximum_price,
    MIN(b.price) AS minimum_price
FROM booking b

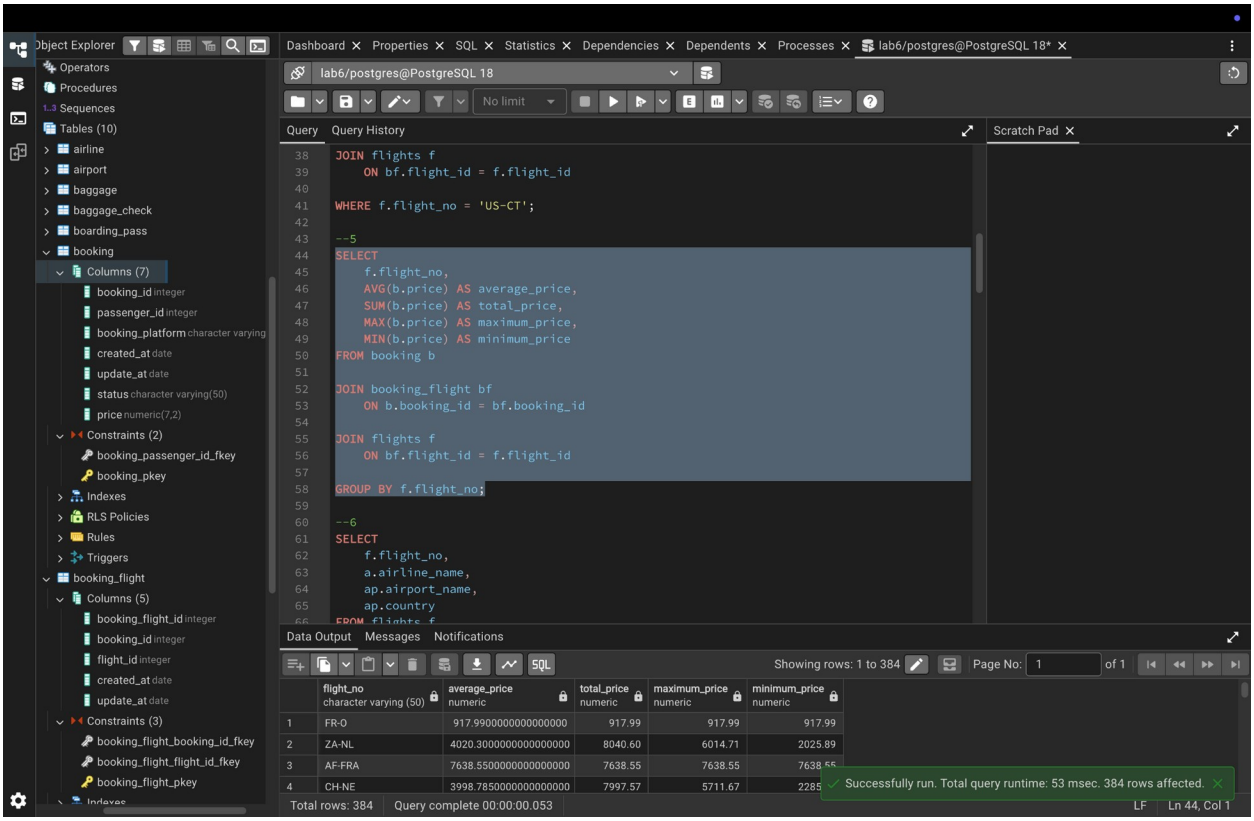
JOIN booking_flight bf
    ON b.booking_id = bf.booking_id

JOIN flights f
    ON bf.flight_id = f.flight_id

GROUP BY f.flight_no;
```

Result: The query was executed successfully. 384 rows were returned.

Screenshot — Task 5



Task 6 — Flights to a specific country

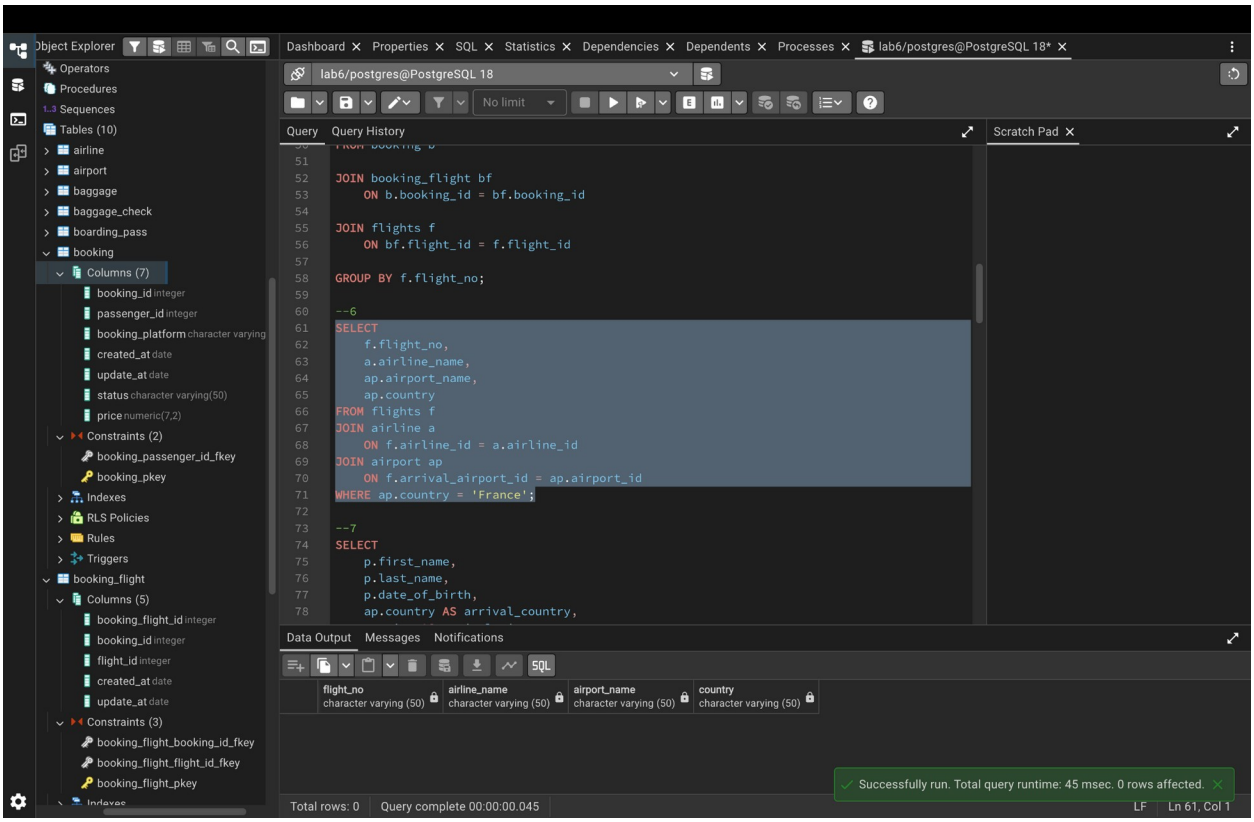
Explanation: The flights, airline and airport tables are combined, and the condition on the arrival airport country selects flights to France.

SQL Query

```
SELECT
    f.flight_no,
    a.airline_name,
    ap.airport_name,
    ap.country
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
JOIN airport ap
    ON f.arrival_airport_id = ap.airport_id
WHERE ap.country = 'France';
```

Result: The query was executed successfully. 0 rows were returned (there are no flights to France in the data).

Screenshot — Task 6



Task 7 — Minor passengers and destinations

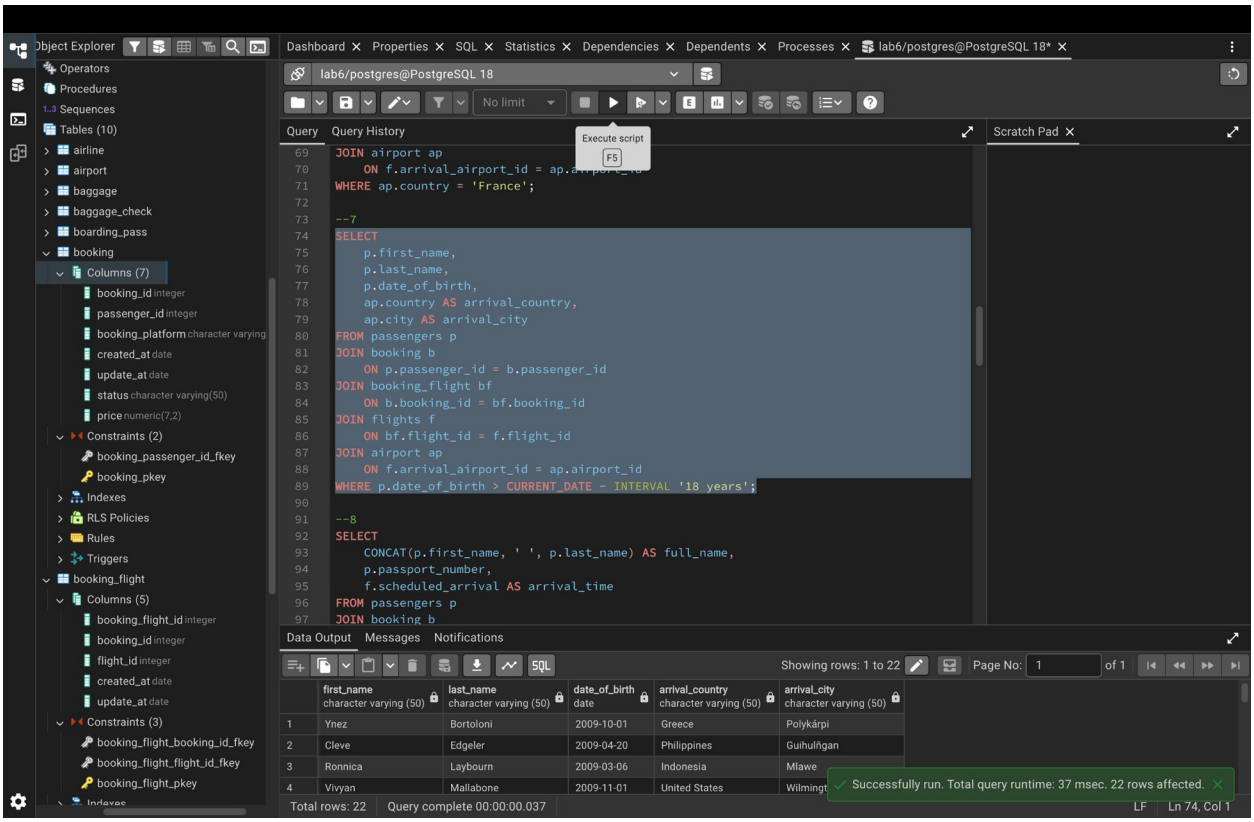
Explanation: Passengers younger than 18 are selected and shown with the country and city of their arrival airport.

SQL Query

```
SELECT
    p.first_name,
    p.last_name,
    p.date_of_birth,
    ap.country AS arrival_country,
    ap.city AS arrival_city
FROM passengers p
JOIN booking b
    ON p.passenger_id = b.passenger_id
JOIN booking_flight bf
    ON b.booking_id = bf.booking_id
JOIN flights f
    ON bf.flight_id = f.flight_id
JOIN airport ap
    ON f.arrival_airport_id = ap.airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
```

Result: The query was executed successfully. 22 rows were returned.

Screenshot — Task 7



Task 8 — Passenger name, passport and arrival time

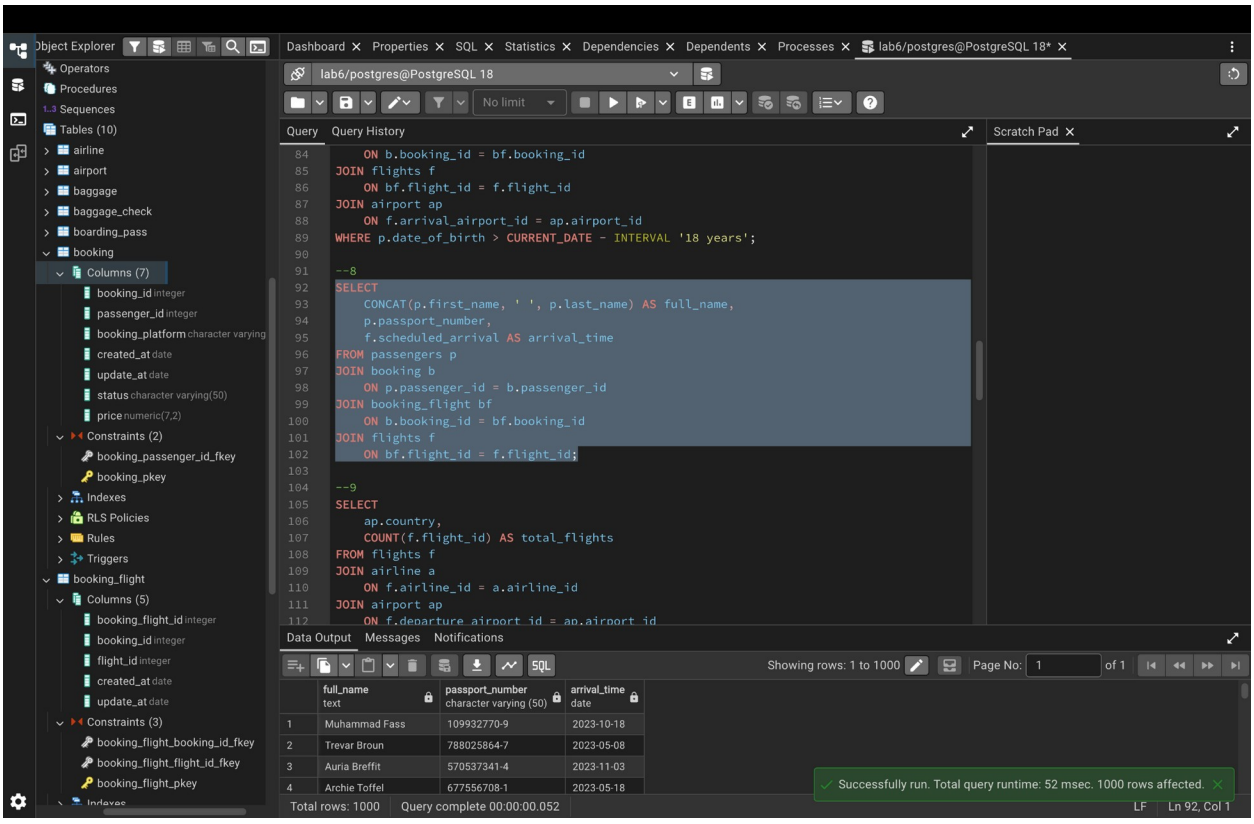
Explanation: The query shows the full name of each passenger, the passport number and the scheduled arrival time of the flight.

SQL Query

```
SELECT
    CONCAT(p.first_name, ' ', p.last_name) AS full_name,
    p.passport_number,
    f.scheduled_arrival AS arrival_time
FROM passengers p
JOIN booking b
    ON p.passenger_id = b.passenger_id
JOIN booking_flight bf
    ON b.booking_id = bf.booking_id
JOIN flights f
    ON bf.flight_id = f.flight_id;
```

Result: The query was executed successfully. 1000 rows were returned.

Screenshot — Task 8



Task 9 — Airline country equals airport country

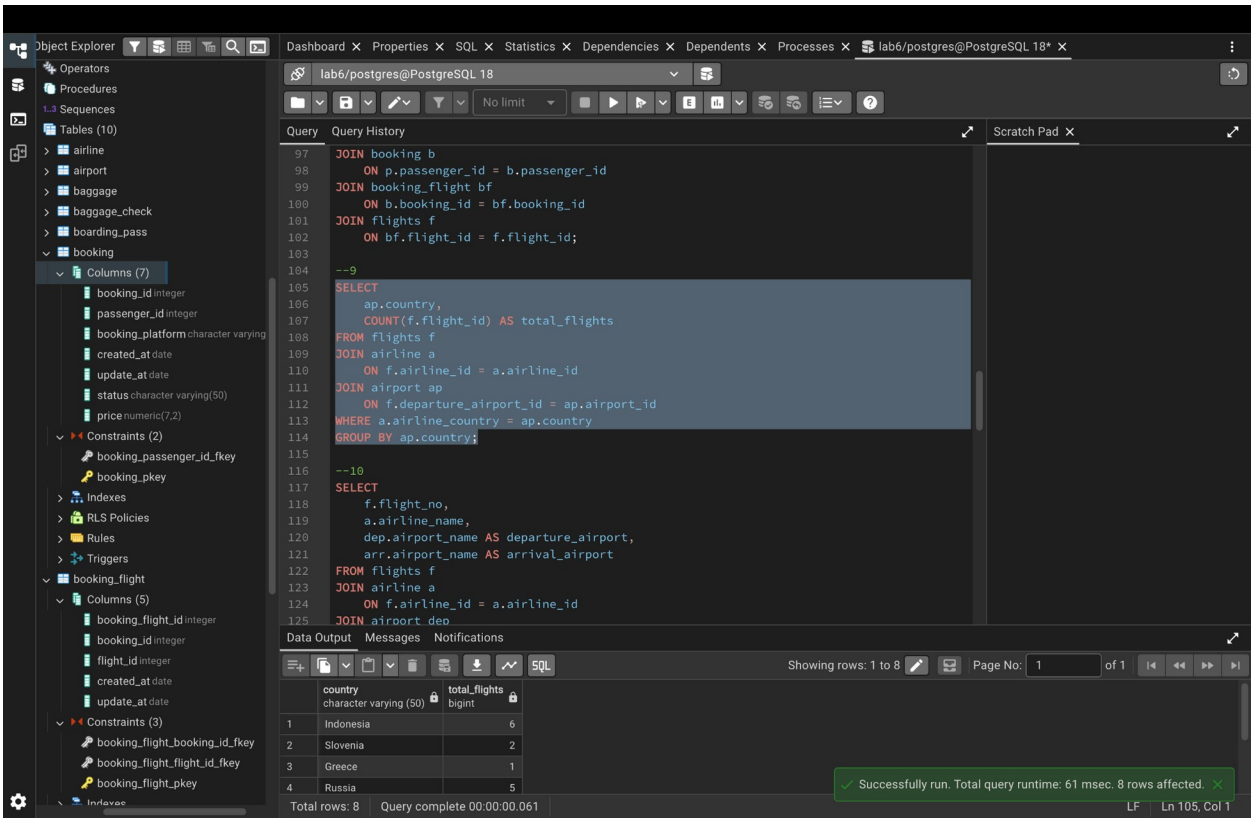
Explanation: Flights are selected where the airline's home country is the same as the departure airport country. The result is grouped by airport country and the flights are counted.

SQL Query

```
SELECT
    ap.country,
    COUNT(f.flight_id) AS total_flights
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
JOIN airport ap
    ON f.departure_airport_id = ap.airport_id
WHERE a.airline_country = ap.country
GROUP BY ap.country;
```

Result: The query was executed successfully. 8 rows were returned.

Screenshot — Task 9



Task 10 — Flights with airline and airports

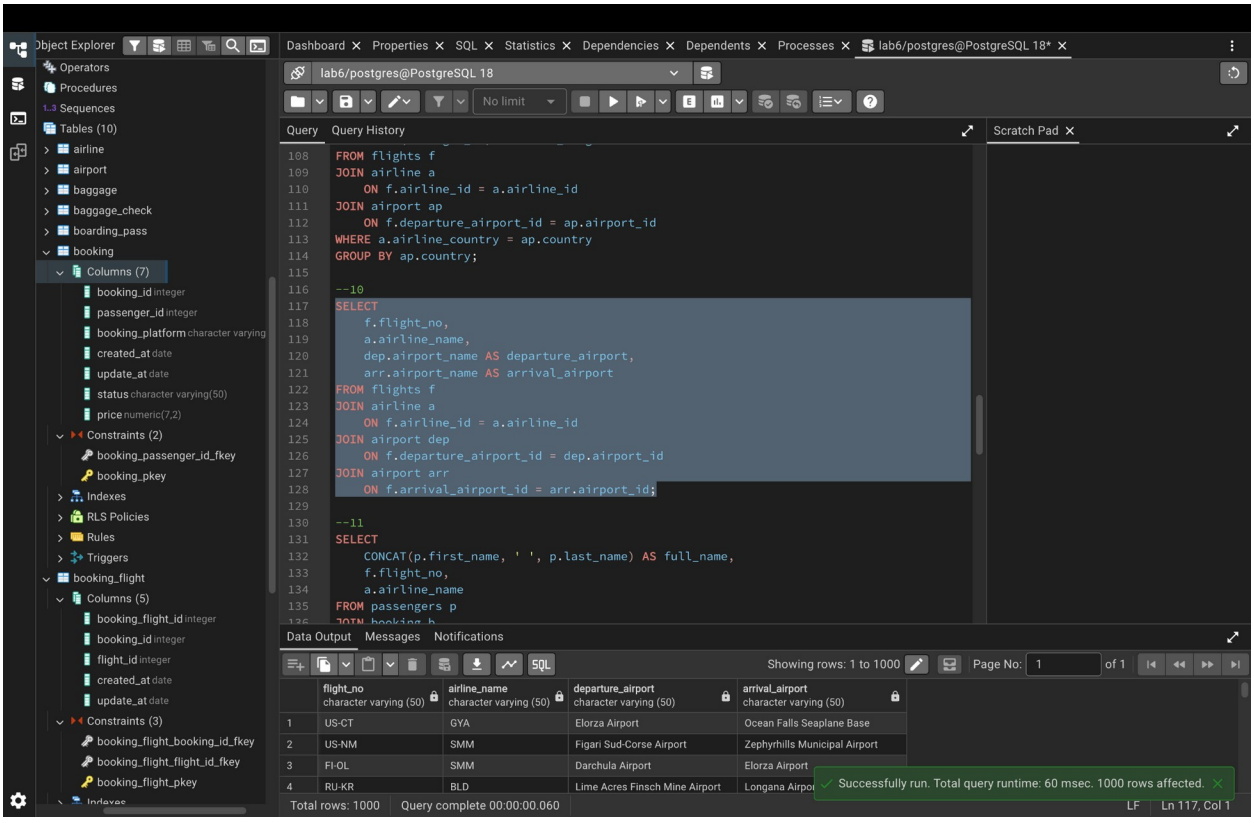
Explanation: The flights table is joined with airline and with the airport table twice (as departure and arrival airport) to show flight number, airline name and both airports.

SQL Query

```
SELECT
    f.flight_no,
    a.airline_name,
    dep.airport_name AS departure_airport,
    arr.airport_name AS arrival_airport
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
JOIN airport dep
    ON f.departure_airport_id = dep.airport_id
JOIN airport arr
    ON f.arrival_airport_id = arr.airport_id;
```

Result: The query was executed successfully. 1000 rows were returned.

Screenshot — Task 10



Task 11 — Passengers of a specific airline

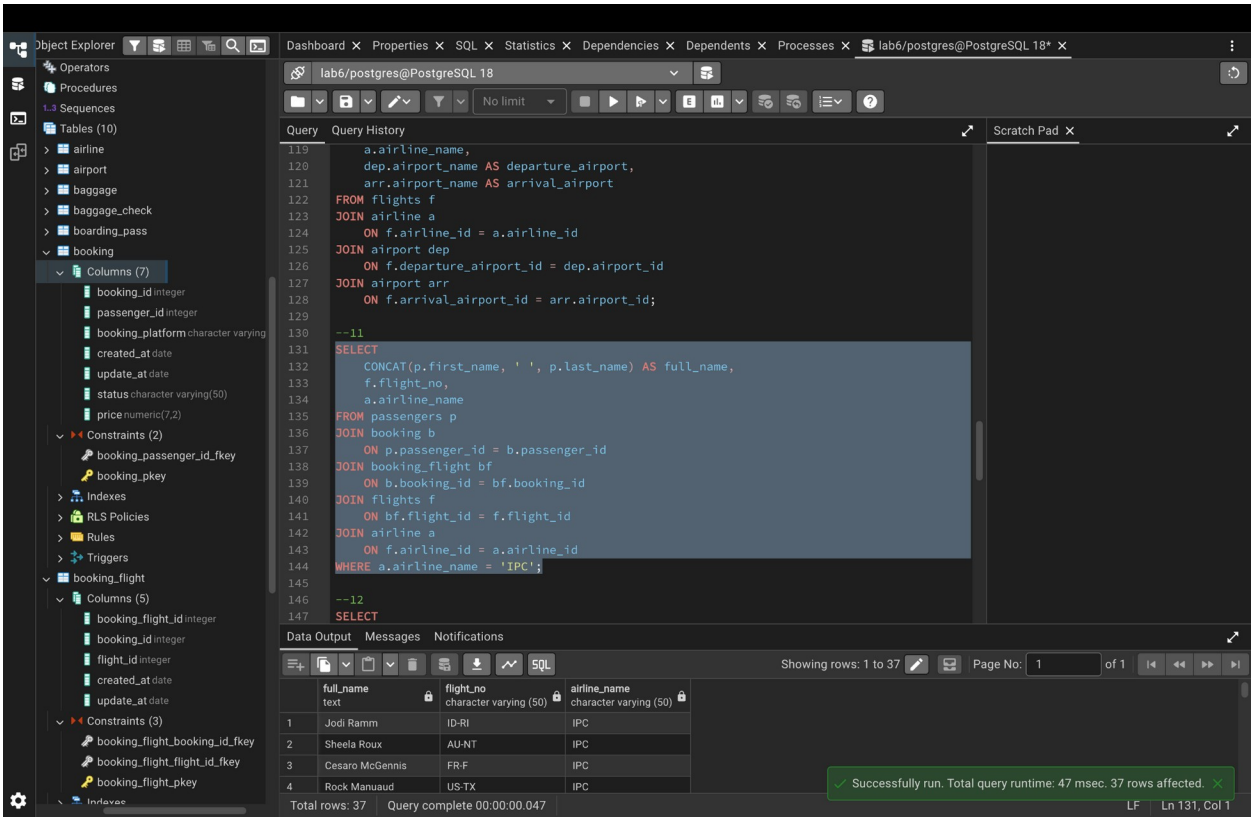
Explanation: All passengers who booked tickets on flights of the airline IPC are shown with their full name, flight number and airline name.

SQL Query

```
SELECT
    CONCAT(p.first_name, ' ', p.last_name) AS full_name,
    f.flight_no,
    a.airline_name
FROM passengers p
JOIN booking b
    ON p.passenger_id = b.passenger_id
JOIN booking_flight bf
    ON b.booking_id = bf.booking_id
JOIN flights f
    ON bf.flight_id = f.flight_id
JOIN airline a
    ON f.airline_id = a.airline_id
WHERE a.airline_name = 'IPC';
```

Result: The query was executed successfully. 37 rows were returned.

Screenshot — Task 11



Task 12 — Passenger count per flight

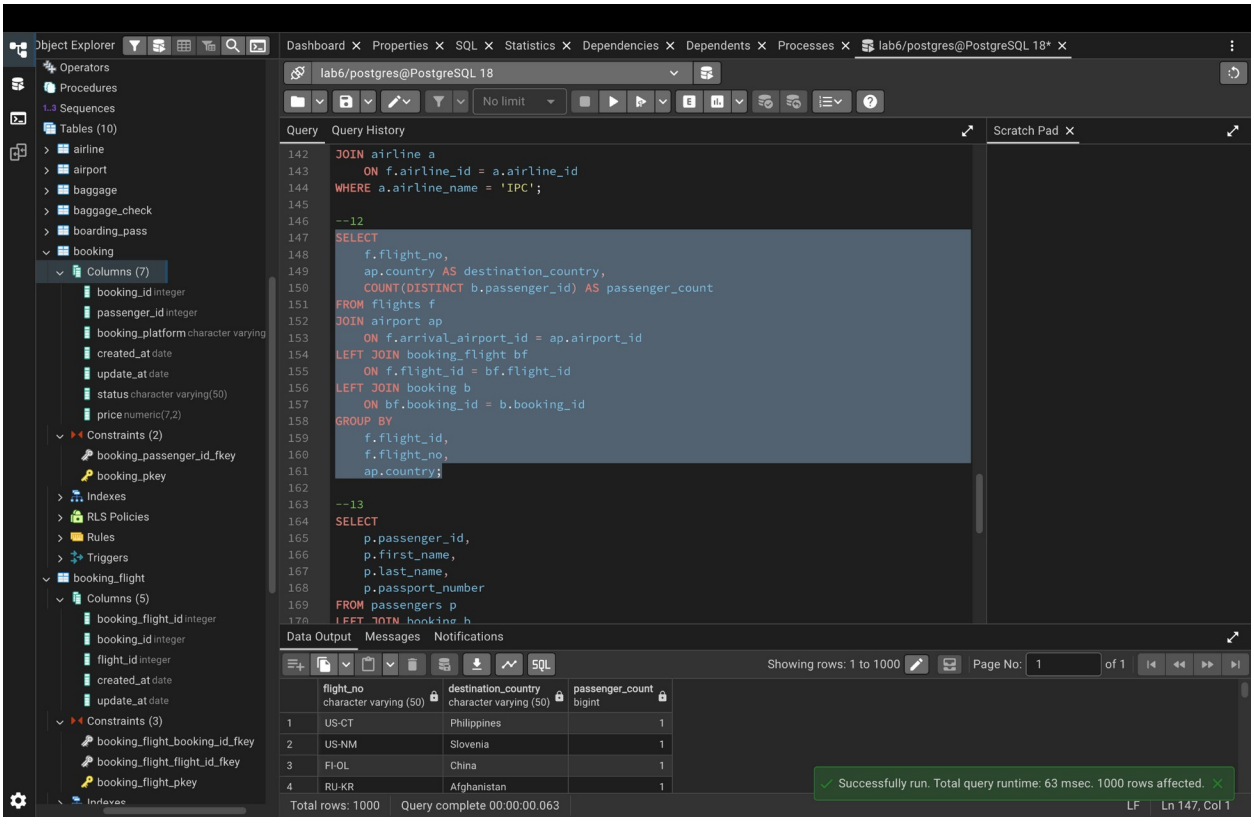
Explanation: For each flight the destination country and the number of booked passengers are shown. LEFT JOIN keeps flights that have no bookings, and their count is 0.

SQL Query

```
SELECT
    f.flight_no,
    ap.country AS destination_country,
    COUNT(DISTINCT b.passenger_id) AS passenger_count
FROM flights f
JOIN airport ap
    ON f.arrival_airport_id = ap.airport_id
LEFT JOIN booking_flight bf
    ON f.flight_id = bf.flight_id
LEFT JOIN booking b
    ON bf.booking_id = b.booking_id
GROUP BY
    f.flight_id,
    f.flight_no,
    ap.country;
```

Result: The query was executed successfully. 1000 rows were returned.

Screenshot — Task 12



Task 13 — Passengers without bookings

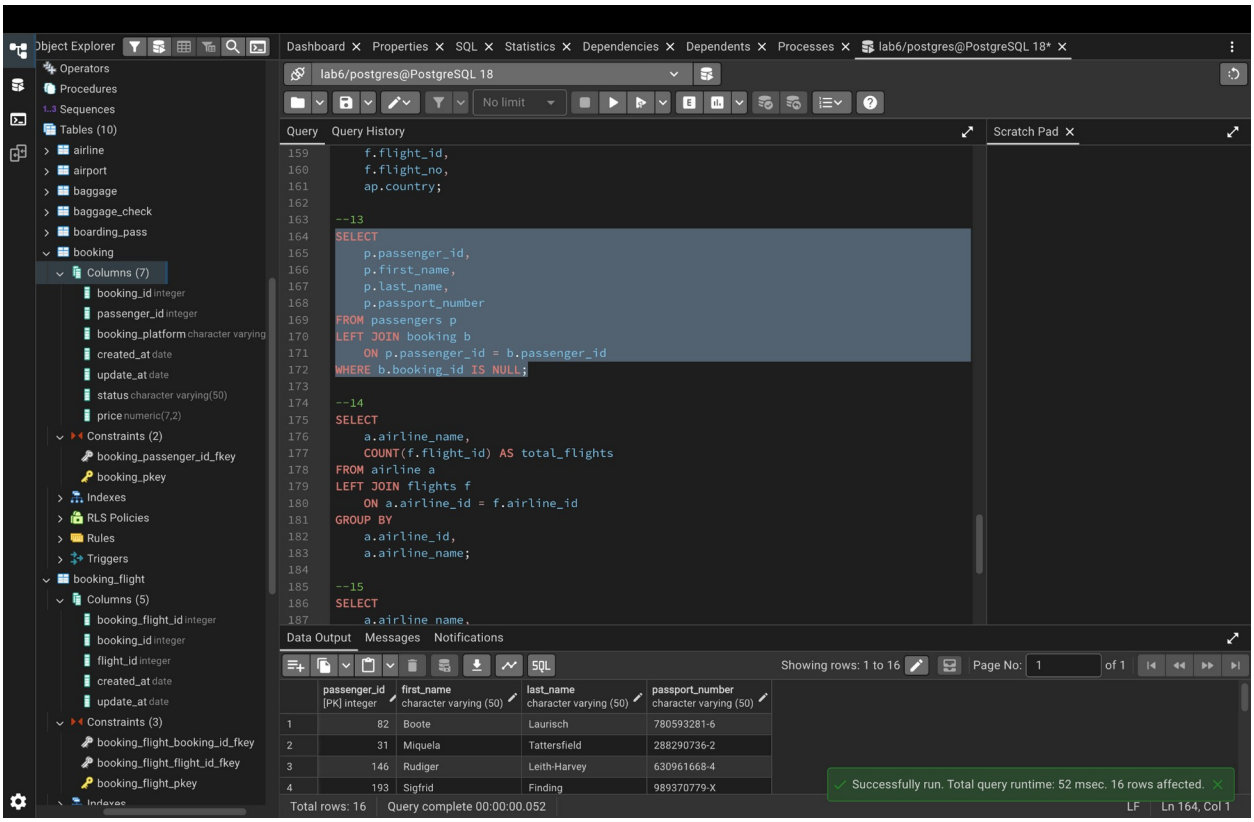
Explanation: A LEFT JOIN between passengers and booking is used. Passengers with no matching booking (b.booking_id IS NULL) have not booked any tickets.

SQL Query

```
SELECT
    p.passenger_id,
    p.first_name,
    p.last_name,
    p.passport_number
FROM passengers p
LEFT JOIN booking b
    ON p.passenger_id = b.passenger_id
WHERE b.booking_id IS NULL;
```

Result: The query was executed successfully. 16 rows were returned.

Screenshot — Task 13



Task 14 — Number of flights per airline

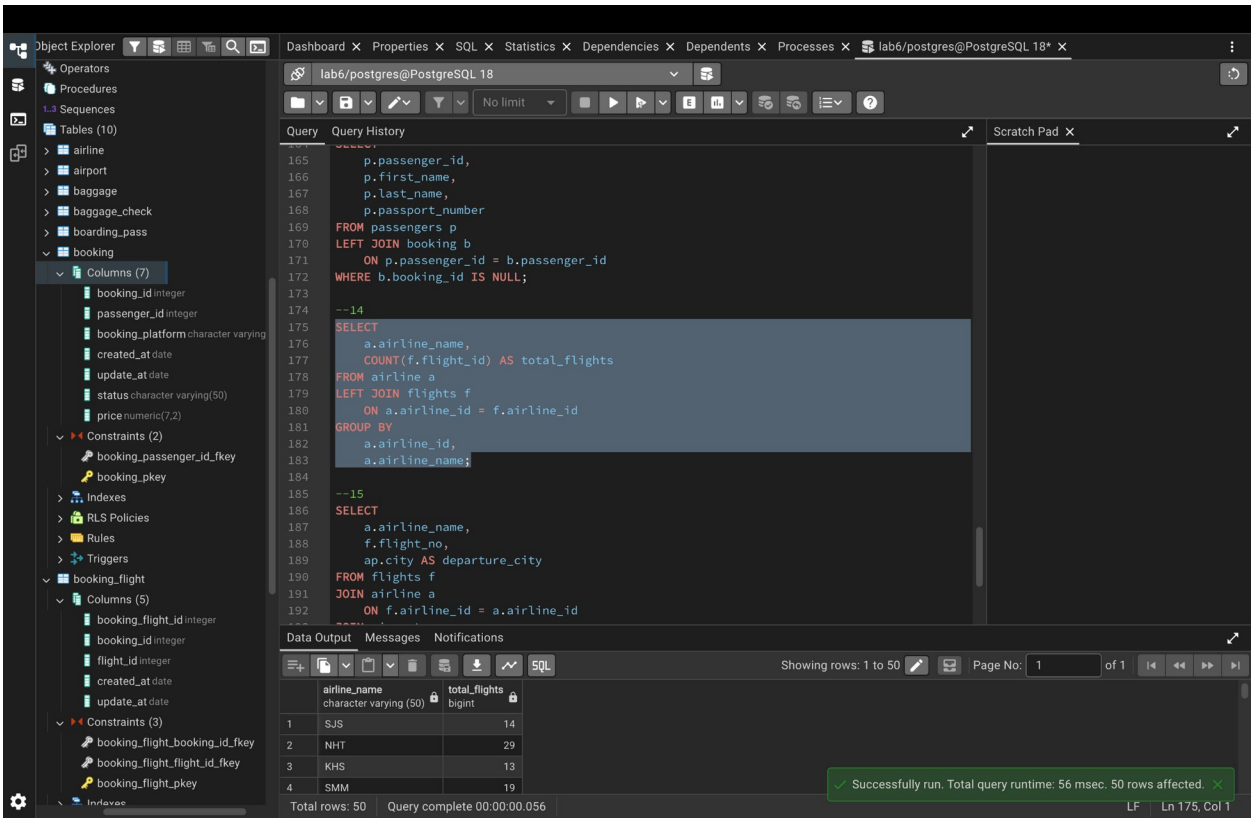
Explanation: Each airline is shown with the total number of its flights. LEFT JOIN includes airlines that have no flights, for which the count is 0.

SQL Query

```
SELECT
    a.airline_name,
    COUNT(f.flight_id) AS total_flights
FROM airline a
LEFT JOIN flights f
    ON a.airline_id = f.airline_id
GROUP BY
    a.airline_id,
    a.airline_name;
```

Result: The query was executed successfully. 50 rows were returned.

Screenshot — Task 14



Task 15 — Flights from a specific country

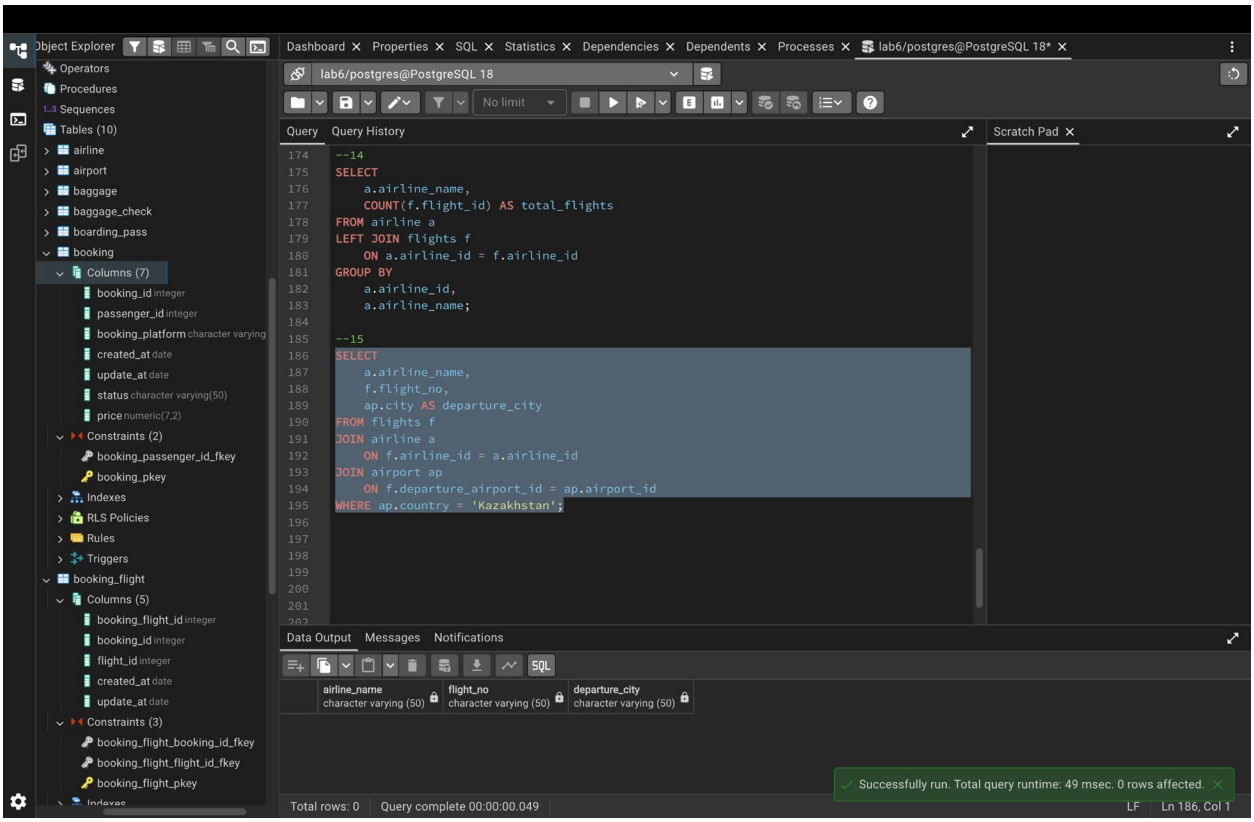
Explanation: Flights departing from airports in Kazakhstan are listed with the airline name, flight number and departure city.

SQL Query

```
SELECT
    a.airline_name,
    f.flight_no,
    ap.city AS departure_city
FROM flights f
JOIN airline a
    ON f.airline_id = a.airline_id
JOIN airport ap
    ON f.departure_airport_id = ap.airport_id
WHERE ap.country = 'Kazakhstan';
```

Result: The query was executed successfully. 0 rows were returned (there are no departures from Kazakhstan in the data).

Screenshot — Task 15



Conclusion

During Laboratory Work 6, JOIN operations were used to combine data from several tables of the airline database. INNER JOIN was used to connect flights, airlines, airports, passengers and bookings. LEFT JOIN was used to find airlines without flights and passengers without bookings, and to keep flights and airlines with zero counts. Aggregate functions (COUNT, AVG, SUM, MAX, MIN) with GROUP BY were used to calculate statistics. The results were checked in pgAdmin after executing the SQL statements.